Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 2

Date: _____ Name: _____

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**35**
Part 2	**30**
CW	**5**
Marks Obtained	

Section A: Multiple Choice Questions (MCQ) (6x1=6 marks)

For each question 1-6, four options are given. Circle the correct option i, ii, iii or iv.

1. To convert a file size from bits to Bytes, what number do you divide by?

i. Divide by 2 **ii.** Divide by 8

iii. Divide by 1,000 **iv.** Divide by 1,024

2. How many Bytes are there in 1 Kibibyte (1 KiB)?

i. 8 Bytes **ii.** 100 Bytes

iii. 1,000 Bytes **iv.** 1,024 Bytes

3. How many Hertz (Hz) are equal to 1 Kilohertz (1 kHz)?

i. 8 Hz **ii.** 100 Hz

iii. 1,000 Hz **iv.** 1,024 Hz

4. In 8-bit ASCII text, how many Bytes are needed to store a word with 12 characters?

i. 6 Bytes **ii.** 12 Bytes

iii. 24 Bytes **iv.** 96 Bytes

5. Which formula calculates the file size (in bits) of a bitmap image?

i. Width + Height + Colour Depth **ii.** Width × Height × Colour Depth

iii. Width × Sample Rate **iv.** Height / 1,024

6. In digital audio recording, how many channels are used in a Stereo sound file?

i. 1 channel **ii.** 2 channels

iii. 4 channels **iv.** 8 channels

Section B: True or False (6x1=6 marks)

7. Write True or False in the box provided for each of the following statements:

No.	Statement	True / False
i.	Exactly 8 bits make up 1 Byte of computer memory.	
ii.	1 Mebibyte (1 MiB) is equal to 1,024 KiB.	
iii.	1 Megahertz (1 MHz) is equal to 1,000 kHz (or 1,000,000 Hz).	
iv.	A Mono audio recording uses 2 separate sound channels.	
v.	Spaces and punctuation marks in a text file also take up 1 Byte (8 bits) each in ASCII.	
vi.	Increasing the colour depth of an image makes its file size smaller.	

Section C: Fill in the Blanks (6x1=6 marks)

8. Fill in the blanks using the correct words from the box below:

Word Bank: [8 | 1,024 | 1,000 | Pixel | Stereo | Hertz (Hz)]

i. To convert a file size from bits to Bytes, we always divide the number of bits by _____.

ii. To convert Bytes into Kibibytes (KiB), we divide the number of Bytes by _____.

iii. A sample rate of 2 kHz is equal to 2 × _____ = 2,000 samples per second.

iv. The smallest single dot of colour in a bitmap image is called a _____.

v. A _____ sound recording uses 2 audio channels (Left and Right).

vi. The unit used to measure sound sample rate (samples per second) is _____.

Section D: Text & Image File Size Calculations (2x3=6 marks)

9. Calculate the file sizes below. Show your step-by-step working clearly:

i. (3m)	**Text File Size:** A plain text story has **2,048 characters** in 8-bit ASCII (1 Byte per character). Find size in **(a) bits**, **(b) Bytes**, and **(c) KiB**: (a) Bits: _____ (b) Bytes: _____ (c) KiB: _____
ii. (3m)	**Image File Size:** A bitmap icon is **64 × 32 pixels** with an **8-bit colour depth**. Find size in **(a) bits**, **(b) Bytes**, and **(c) KiB**: (a) Bits: _____ (b) Bytes: _____ (c) KiB: _____

Section E: Sound File Size Calculations (2x3=6 marks)

10. Use (Sample Rate × Resolution × Time × Channels) to calculate the audio file sizes:

i. (3m)	**Mono Sound File:** Sample Rate = **1,000 Hz**, Resolution = **8 bits**, Time = **8 seconds**, **Mono (1 channel)**. Find size in **(a) bits** and **(b) Bytes**: (a) Size in bits: _____ (b) Size in Bytes: _____
ii. (3m)	**Stereo Sound File:** Sample Rate = **1,024 Hz**, Resolution = **16 bits**, Time = **4 seconds**, **Stereo (2 channels)**. Find size in **(a) Bytes** and **(b) KiB**: (a) Size in Bytes: _____ (b) Size in KiB: _____